

0 4 . 1

Name **three** structures that are found in **some** prokaryotic cells but are **not** found in **all** prokaryotic cells.

[2 marks]

1 _____

2 _____

3 _____

0 4 . 2

Table 3 shows the resolution of three types of microscope.

Table 3

Microscope	Resolution / μm
Optical	0.25
Scanning electron (SEM)	0.02
Transmission electron (TEM)	0.0002

Table 4 shows information about yeast cells and mitochondria.

Table 4

	Mean diameter / μm
Yeast cells	3.0
Mitochondria	0.22

Identify the correct microscope to use in each of the following investigations.

Use your knowledge of microscopes and information in **Table 3** and **Table 4**.

Give **two** reasons for each choice of microscope.

[4 marks]

Investigation 1 – to study the living cells in a yeast culture that contains both living cells and dead cells. The living cells remain colourless and the dead cells are stained blue when a dye is added.

Type of microscope _____

Reason 1 _____

Reason 2 _____

Investigation 2 – to observe the three-dimensional structure of mitochondria.

Type of microscope _____

Reason 1 _____

Reason 2 _____

Turn over for the next question

Turn over ►

